

CS1812

DISCRETE MATHEMATICS & SET THEORY

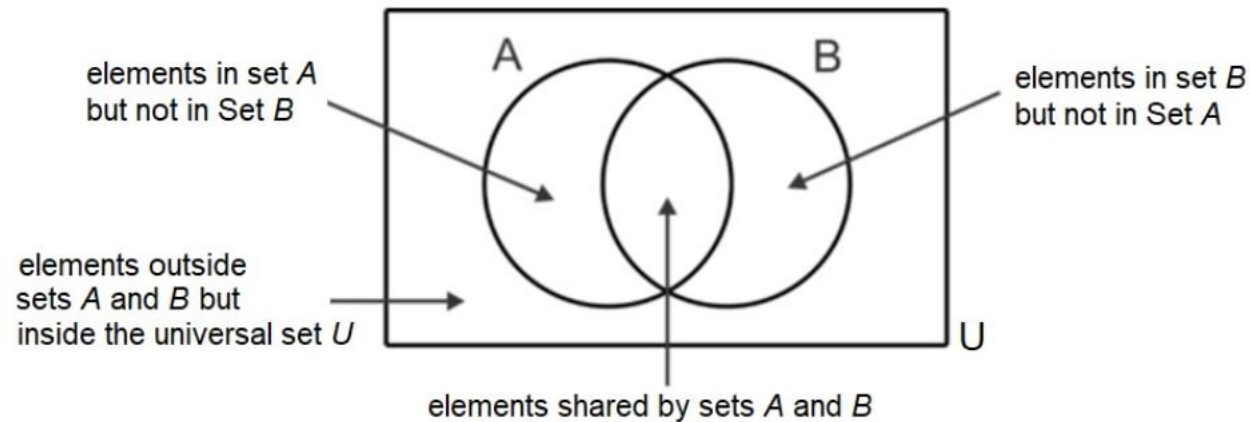


Module 1 : Sets

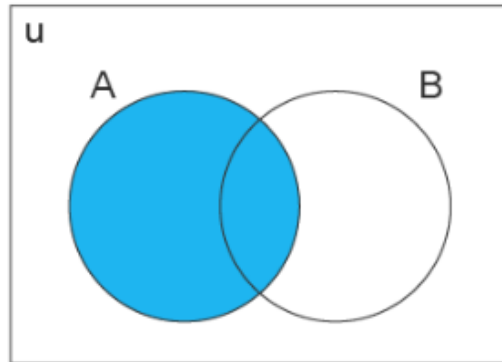
- Venn diagram
- Operations on sets
- Basic Laws

Venn diagram

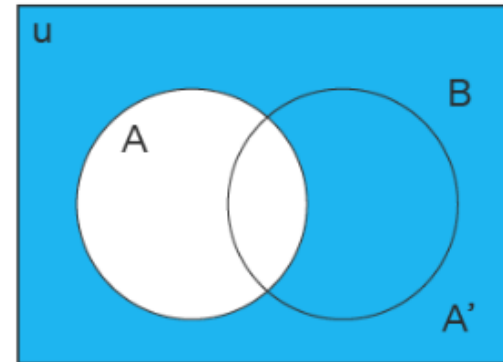
In set theory, a Venn diagram is a visual way to show the relationships between different sets. Each set is shown as a circle. The universal set (U) is usually represented by a rectangle around all the circles, and it contains all possible elements being considered.



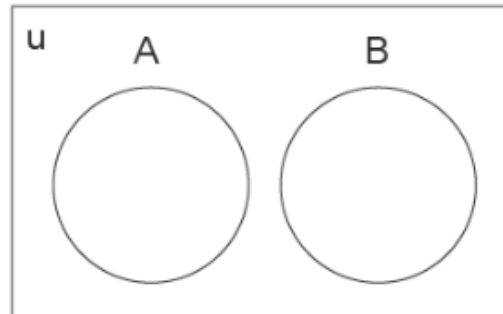
Venn diagram



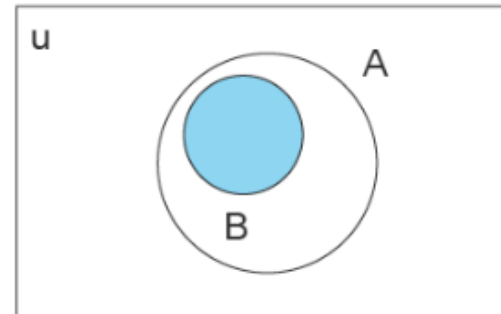
Set A



A' is the complement of A

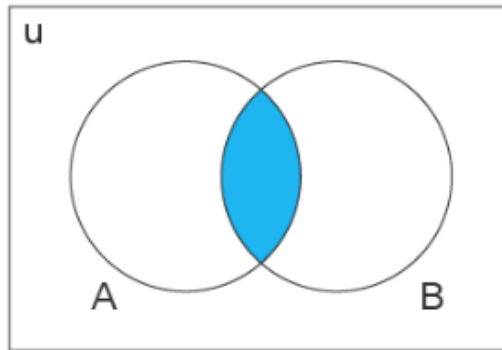


A and B are disjoint sets



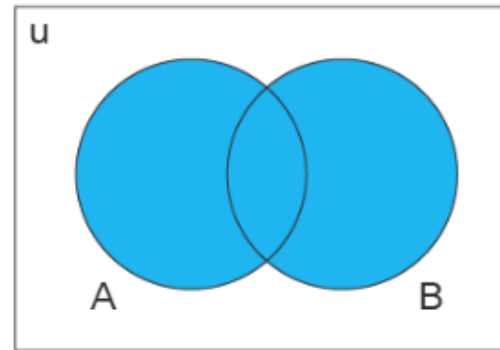
$B \subset A$
B is proper subset of A

Venn diagram



$$A \cap B$$

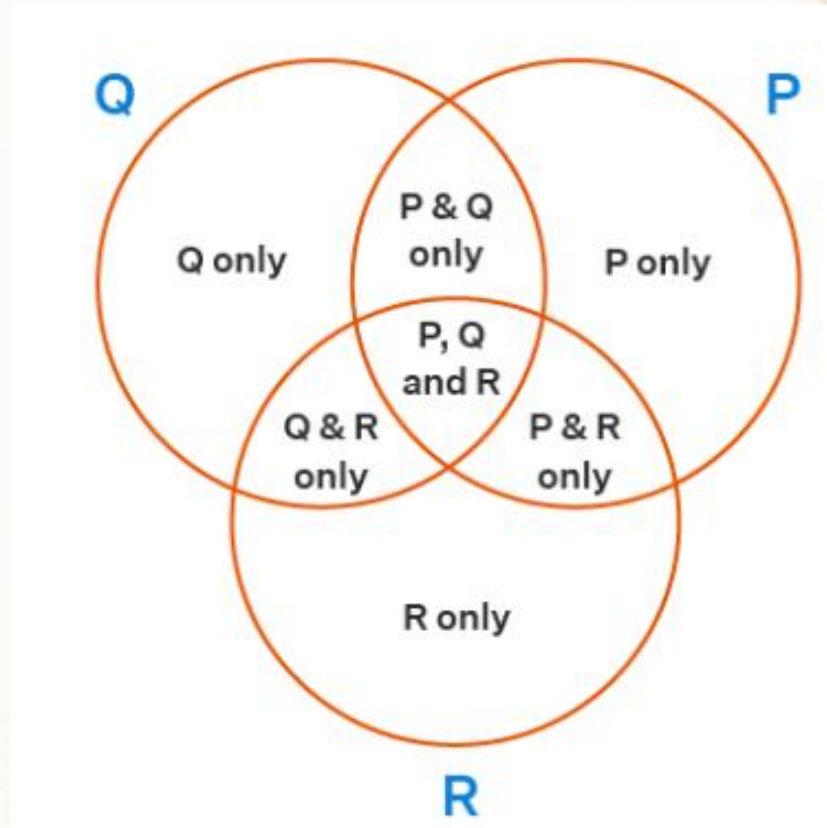
A and B are overlapping sets



$$A \cup B$$

A and B are overlapping sets

Venn diagram



Operations on sets

There are five kinds of set operations which are as follows :

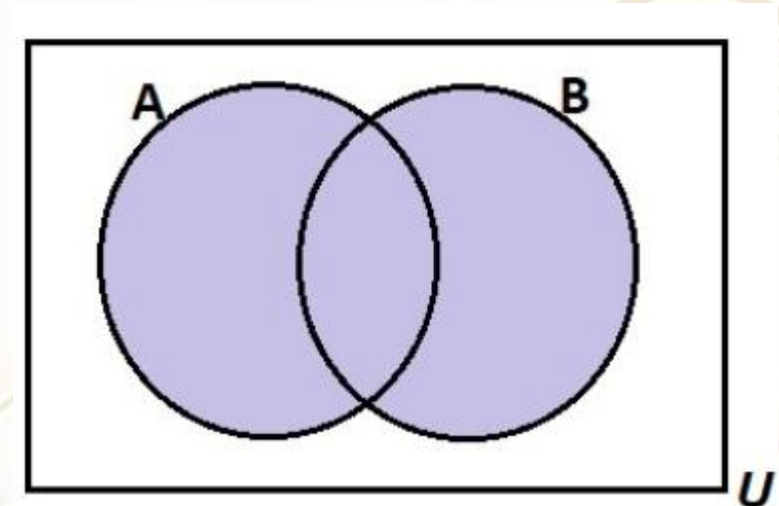
- Union
- Intersection
- Complement
- Difference
- Symmetric difference

Union of sets

The **union** of two sets contains all the elements contained in either (or both) sets. Union of two sets A and B is denoted by $A \cup B$. The shaded region represents $A \cup B$. Also, if $x \in A \cup B$, then $x \in A$ or $x \in B$ (or both).

Ex : $A = \{4, 5, 6\}$, $B = \{6, 7\}$.

Then $A \cup B = \{4, 5, 6, 7\}$

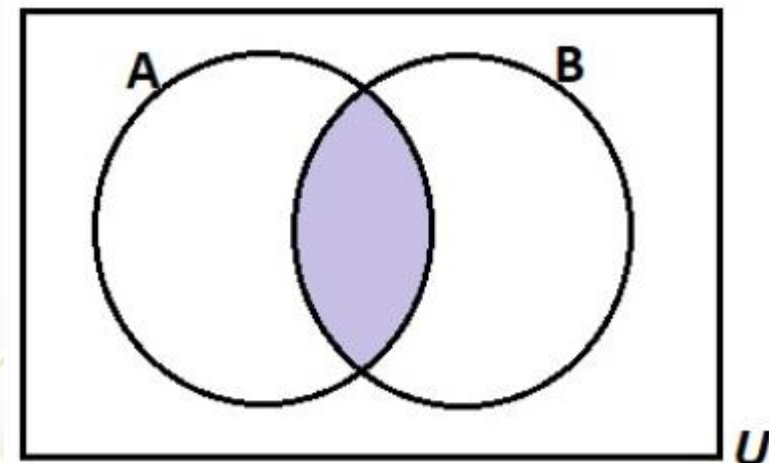


Intersection of sets

The **intersection** of two sets contains only the elements that are in both sets. Intersection of two sets A and B is denoted by $A \cap B$. The shaded region represents $A \cap B$. Also, if $x \in A \cap B$, then $x \in A$ and $x \in B$.

Ex : $A = \{4, 5, 6\}$, $B = \{6, 7\}$.

Then $A \cap B = \{6\}$



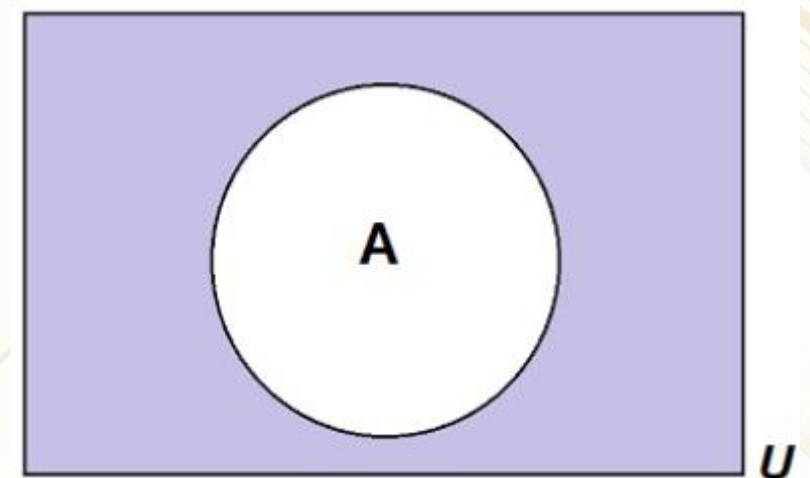
Complement of a set

The **complement** of a set contains everything that is not in the set but still within the universal set. If A is a set, its complement is denoted by A' or A^c and is given by $A' = U - A$. The shaded region represents A' .

If $x \in A'$, then $x \in U$ and $x \notin A$.

Ex : Let $U = \{1, 2, 3, 4, 5, 6\}$ and $A = \{2, 4\}$.

Then $A' = U - A = \{1, 3, 5, 6\}$



Difference of two sets

The **difference** of the sets A and B in this order is the set of elements which belong to A but not to B. It is denoted by $A - B$.

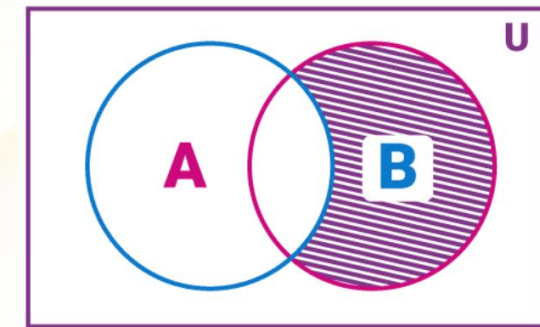
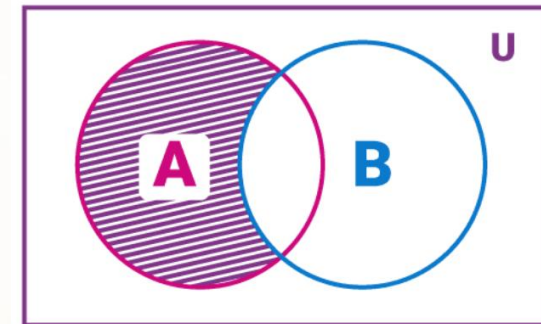
The shaded region represents $A - B$ and $B - A$.

If $x \in A - B$, then $x \in A$ and $x \notin B$.

Similarly, If $x \in B - A$, then $x \in B$ and $x \notin A$.

Ex : $A = \{2, 4\}$, $B = \{4, 5, 6, 7\}$

Then $A - B = \{2\}$ and $B - A = \{5, 6, 7\}$



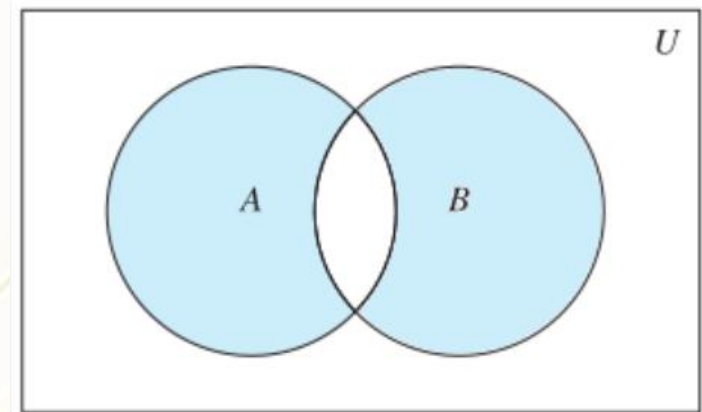
Symmetric difference

The set which contains the elements which are either in set A or in set B but not in both is called the **symmetric difference** between two given sets. It is denoted by $A \Delta B$ and $A \Delta B = (A - B) \cup (B - A)$. The shaded region represents $A \Delta B$.

Ex : Let $A = \{1, 2, 3, 4, 5, 6\}$ and $B = \{2, 4, 6, 8, 10\}$.

Then $A - B = \{1, 3, 5\}$ and $B - A = \{8, 10\}$

Thus $A \Delta B = (A - B) \cup (B - A) = \{1, 3, 5, 8, 10\}$



Basic Laws

$A \cup \emptyset = A$ $A \cap U = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$(A')' = A$ $A \cup A' = U$ $A \cap A' = \emptyset$ $\emptyset' = U$ $U' = \emptyset$	Complementation laws

Basic Laws

	$(A \cup B)' = A' \cap B'$ $(A \cap B)' = A' \cup B'$	De Morgan's Laws
	$A - (B \cup C) = (A - B) \cap (A - C)$ $A - (B \cap C) = (A - B) \cup (A - C)$	De Morgan's Law for the difference of sets ○
	$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
	$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associate laws

Basic Laws

$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A - B = A \cap B'$	Set difference law